Video links

https://www.youtube.com/watch?v=QHPYRwCWAzM

https://www.youtube.com/watch?v=cQPVXrV0GNA

Further reference
https://www.savemyexams.com/igcse/biology/cie/23/revision-notes/1-characteristics-and-classification-of-living-organisms/1-1-characteristics-classification-and-features-of-organisms/1-1-1-characteristics-of-living-organisms/

Characteristics of Living Things

Living organisms are things that have life, such as plants, animals, humans, fungi and bacteria. Although living organisms may differ in their size, shape and structure, they share certain characteristics that distinguish them from non-living things.

Scientists use these characteristics to identify living organisms and understand how they survive, grow and interact with their environment.

The acronym MRS GREN+ H helps us remember the eight characteristics of living things.



M – Movement

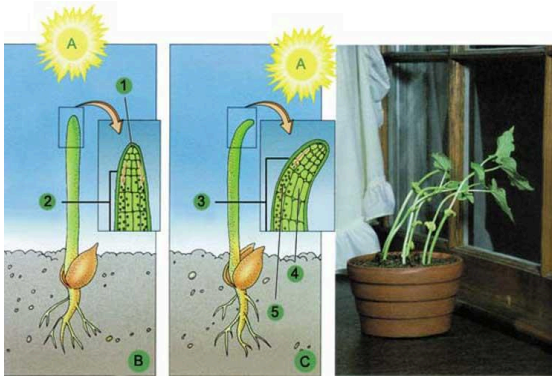
- **Definition**: The ability of an organism to change its position or move part of its body.

Movement is the ability of an organism or part of an organism to change its position.

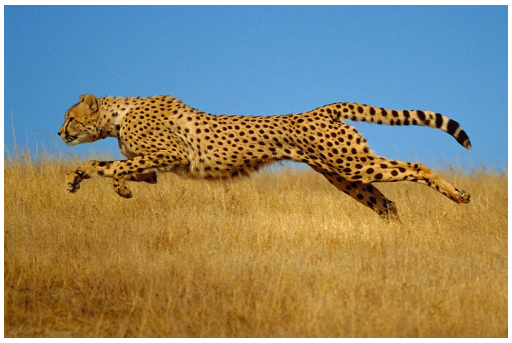
- Animals move from one place to another to find food, shelter or mates.
- Plants also show movement, although it is usually much slower.

- Example: A sunflower shoot grows towards light, while roots grow towards gravity.

Why it matters: Movement helps organisms find food, escape danger, and respond to their environment.



R –



Respiration

Respiration is the chemical process by which living cells release energy from food.

- Energy is needed for growth, movement, repair and other life processes.

Types:

  - **Aerobic respiration** (uses oxygen):
    `Glucose + Oxygen → Carbon dioxide + Water + Energy`

  - **Anaerobic respiration** (without oxygen): common in some microorganisms and muscle cells under stress.
- Respiration occurs in both plants and animals, day and night.

Remember: Breathing and respiration are not the same. Breathing is the movement of air into and out of the lungs, while cellular respiration releases energy inside cells.

Why it matters: Energy released during respiration powers all life processes.

S – Sensitivity (Response to Stimuli)

Sensitivity is the ability of living organisms to detect and respond to changes (stimuli) in their surroundings.

- **Humans pull their hands away from hot objects.**
- **Pupils become smaller in bright light.**
- **The leaves of the touch-me-not plant fold when touched.**

Why it matters: Helps organisms survive by adapting to their surroundings.

G - 

Growth

Growth is a permanent increase in the size and dry mass of an organism, often through an increase in cell number or cell size.

- **A seed develops into a seedling and eventually a mature plant.**
- **A baby grows into an adult.**
- **Growth requires energy and nutrients.**

- **Why it matters**: Growth is essential for development and the continuation of life cycles.



R – Reproduction

- **Definition**: The process by which living things produce offspring (new individuals).

- **Types**:
    - **Asexual reproduction**: One parent, offspring are genetically identical (e.g., bacteria).
    - **Sexual reproduction**: Two parents, offspring are genetically varied (e.g., humans, animals).
- **Why it matters**: Ensures the survival of a species.

E – Excretion

Excretion is the removal of metabolic waste products and substances present in excess from the body.

- **Kidneys remove urea, excess water and salts through urine.**
- **Lungs remove carbon dioxide and water vapour.**
- **Plants can release excess water and gaseous waste products.**

Important distinction: Excretion removes metabolic waste, whereas egestion removes undigested food from the digestive tract

- **Why it matters**: Waste build-up can be toxic; excretion maintains internal balance.

N – Nutrition

- **Definition**: The process of taking in food and using it for growth and energy.
- **Types**:
    - **Autotrophs**: Make their own food (e.g., plants via photosynthesis).
    - **Heterotrophs**: Depend on other organisms for food (e.g., humans, animals).
- **Why it matters**: Provides materials for energy, growth, and repair.

H – Homeostasis

Homeostasis is the maintenance of a relatively stable internal environment despite changes inside or outside the organism.

Examples include:

- Sweating when body temperature rises.
- Shivering when body temperature falls.
- Regulating blood glucose concentration.
- Maintaining water balance in the body.
    -
- **Why it matters**: Keeps conditions constant for enzyme activity and life functions.

Summary Table

Characteristic	Description	Example
Movement	Changing position or internal motion	Running, sunflower turning
Respiration	Releasing energy from food	Breathing, cellular respiration
Sensitivity	Detecting and responding to stimuli	Eyes blinking in bright light
Growth	Increase in size or complexity	Child growing taller
Reproduction	Producing offspring	Birds laying eggs
Excretion	Removal of waste	Sweating, urinating
Nutrition	Getting or making food	Eating, photosynthesis
Homeostasis	Keeping internal conditions stable	Sweating to cool the body

Application-Based Questions

1. A student observes that a plant bends towards a window. Identify the characteristics of life demonstrated and explain its importance.
2. A car moves, uses fuel and releases exhaust gases. Does this mean a car is alive? Justify your answer using characteristics of living organisms.
3. Explain why a seed can be considered living even when it does not appear to grow.
4. Compare respiration and breathing using one scientific difference.
5. Explain how sweating helps maintain homeostasis.
6. A mule cannot normally reproduce. Explain why it is still classified as a living organism

1. Which process releases energy from food inside cells?

Breathing

Respiration

Nutrition

2. A touch-me-not plant closes its leaves when touched. Which characteristic is shown?

Sensitivity

Reproduction

Excretion

3. Why is a car not considered a living organism?

It cannot move

It is not made of cells and cannot perform all life processes

It uses fuel

4. Which example demonstrates homeostasis?

A bird laying eggs

A person sweating on a hot day

A seed germinating